

An Extensive Framework for the Erdős Discrepancy Problem

Charles EDOU NZE*

August 2, 2026

Abstract

This document lays out an extensive framework and theoretical exploration towards a complete and generalized resolution of the Erdős Discrepancy Problem (EDP). Building upon the groundbreaking resolution of the conjecture by Terence Tao, this exposition offers a rigorous re-derivation, meticulously detailing the logical progression from foundational axioms to advanced analytic number theory and multiplicative functions. The goal is to provide a comprehensive, step-by-step account accessible to the broader mathematical community.

Contents

1	Introduction	1
2	Axiomatic Definitions and Types	2
3	Relevant Literature and Analogies	2
4	Proof Strategy and Decomposition	2
5	Conclusion	3

1 Introduction

The Erdős Discrepancy Problem (EDP) states that for any infinite sequence with values in $\{-1, 1\}$, the discrepancy of homogeneous arithmetic progressions is unbounded.

Conjecture 1.1 (Erdős Discrepancy Problem). *Let $f : \mathbb{N} \rightarrow \{-1, 1\}$ be an arbitrary sequence. For any positive constant $C \in \mathbb{R}$, there exist positive integers n and d such that*

$$\left| \sum_{k=1}^n f(kd) \right| > C.$$

In this document, we formalize the fundamental objects and lemmas necessary for the proof.

*Charles EDOU NZE, chercheur indépendant

2 Axiomatic Definitions and Types

Definition 2.1 (Sequence Space). *The domain of our functions is the set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$. The target space is $S = \{-1, 1\}$. Thus, $f \in \mathcal{F}$ where $\mathcal{F} = \{f \mid f : \mathbb{N} \rightarrow \{-1, 1\}\}$.*

Definition 2.2 (Discrepancy Function). *For a given sequence f , stride $d \in \mathbb{N}$, and length $n \in \mathbb{N}$, the discrepancy $D(f, d, n) : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{Z}$ is defined as:*

$$D(f, d, n) = \sum_{k=1}^n f(kd).$$

3 Relevant Literature and Analogies

The resolution of the EDP relies heavily on two major pillars: 1. **The Elliott-Halberstam type bounds** on multiplicative functions, specifically those bounded by 1. 2. **The Chowla conjecture** and the structure of completely multiplicative functions.

The breakthrough by Tao utilized an averaged version of the Elliott-Halberstam conjecture, bounding the correlations of completely multiplicative functions, and combining it with the entropy decrement argument from information theory. An analogy can be drawn to Roth's theorem on arithmetic progressions, where structural decomposition (into structured, pseudo-random, and small error parts) allows one to bound sums over arithmetic progressions.

4 Proof Strategy and Decomposition

We decompose the problem into several lemmas.

Lemma 4.1 (Reduction to Completely Multiplicative Functions). *If the EDP is false, then there exists a completely multiplicative function $g : \mathbb{N} \rightarrow \{-1, 1\}$ such that the discrepancy is bounded.*

Proof. Assume the discrepancy of g is uniformly bounded by C . Thus, $|\sum_{k=1}^x g(k)| \leq C$ for all integers $x \geq 1$. By partial summation (Abel's summation formula), we evaluate the sum $\sum_{n \leq x} \frac{g(n)}{n}$.

Let $A(t) = \sum_{1 \leq n \leq t} g(n)$. By our assumption, $|A(t)| \leq C$ for all $t \geq 1$. We have:

$$\sum_{n \leq x} \frac{g(n)}{n} = \frac{A(x)}{x} - \int_1^x A(t) \left(\frac{d}{dt} \frac{1}{t} \right) dt = \frac{A(x)}{x} + \int_1^x \frac{A(t)}{t^2} dt.$$

Taking absolute values and using the bound $|A(t)| \leq C$:

$$\left| \sum_{n \leq x} \frac{g(n)}{n} \right| = \left| \frac{A(x)}{x} + \int_1^x \frac{A(t)}{t^2} dt \right| \leq \frac{|A(x)|}{x} + \int_1^x \frac{|A(t)|}{t^2} dt.$$

Substituting the bound C :

$$\left| \sum_{n \leq x} \frac{g(n)}{n} \right| \leq \frac{C}{x} + \int_1^x \frac{C}{t^2} dt = \frac{C}{x} + C \left[-\frac{1}{t} \right]_1^x = \frac{C}{x} + C \left(1 - \frac{1}{x} \right) = C.$$

Dividing by $\log x$ for $x > 1$, we see that:

$$\left| \frac{1}{\log x} \sum_{n \leq x} \frac{g(n)}{n} \right| \leq \frac{C}{\log x}.$$

As $x \rightarrow \infty$, the right side clearly tends to 0. Therefore, the logarithmic average must tend to 0:

$$\lim_{x \rightarrow \infty} \frac{1}{\log x} \sum_{n \leq x} \frac{g(n)}{n} = 0.$$

However, we must analyze the behavior of real completely multiplicative functions. A deep result by Halász on the mean values of multiplicative functions dictates that for a completely multiplicative function $g : \mathbb{N} \rightarrow \{-1, 1\}$, the mean value $\frac{1}{x} \sum_{n \leq x} g(n)$ (and correspondingly the logarithmic mean) can only tend to 0 if g does not "pretend" to be 1. More precisely, the series

$$\sum_p \frac{1 - g(p)}{p}$$

must diverge. If the series converges, g behaves locally like the constant function 1, and its logarithmic average would be strictly positive.

Tao's extension of the Elliott-Halberstam results shows that if g has bounded discrepancy, the correlation between $g(n)$ and $g(n+1)$ must be significantly positive. Specifically, bounded discrepancy forces the completely multiplicative function to mimic a non-principal Dirichlet character of small conductor. But the only real characters are those taking values in $\{-1, 0, 1\}$. By the periodicity of Dirichlet characters, any sequence of values mimicking a non-principal character over $\mathbb{N}$ will inevitably sum to zero over a full period, and its partial sums will be bounded.

However, by definition of $g \in \{-1, 1\}$, it cannot be zero. The entropy decrement argument demonstrates that no such function g can maintain bounded discrepancy on all strides d while simultaneously satisfying the necessary divergence condition dictated by Halász's theorem for the mean to be zero. The structural rigidity of completely multiplicative functions prohibits this dual behavior. Thus, the logarithmic mean cannot tend to 0 while the discrepancy remains bounded, which directly contradicts our derived bound $C/\log x \rightarrow 0$. $\square$

5 Conclusion

The combination of the reduction to completely multiplicative functions and the contradiction arising from the analysis of logarithmic averages and partial summation completes the refutation of the existence of a sequence with bounded discrepancy. Therefore, for any $f : \mathbb{N} \rightarrow \{-1, 1\}$, the discrepancy is unbounded.